AKAASH GARG

Full-Stack Systems Architect · Real-Time & Distributed Systems · Unity · Photon Fusion 2 · Node · PostgreSQL · 17+ yrs

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PROFILE

Seventeen years shipping software — twelve running my own agency across web, mobile and backend delivery, then six against the tightest constraint in commercial software: a standalone headset at **72fps, 13.9ms a frame**, with the server still holding authority. I design the whole stack of a real-time product and then build it — where authority lives and what crosses the wire, the PostgreSQL schemas and APIs underneath, offline-first ingestion that survives a venue with no reliable internet, and the client that has to render it on budget.

Founder of **2 Dude Studios**, leading six people across an **8-player co-located VR arena**, a **multi-tenant training platform** on PostgreSQL, and a survival game with a public Steam demo. Earlier: a Unity VR client bolted to a Java backend nobody was allowed to change — **sub-100ms**, 50M+ user platform, zero critical defects in three months.

EXPERIENCE

Founder & Lead XR Engineer · 2 Dude Studios

Jul 2023 – Present · Remote

XR studio I founded — 6-person delivery team across location-based VR, enterprise training and games. Steam-published developer.

- **Architected the free-roam LBE VR arena — 8 co-located players on standalone Quest 3S.** The venue operator's PC is the single authoritative host owning all session state; headsets contribute input only. A printed floor mat is the shared coordinate frame, so nothing spatial crosses the wire and Fusion replicates only poses already in a common frame. Depth-sensor floor correction; offline-first telemetry into Supabase with disk queueing, backoff and crash-journal reconciliation, behind a React operator dashboard.
- **Shipped *Photon Enemy AI* v1.0.0, a commercial Unity package** — finite-state-machine enemies running two runtimes from one codebase: networked on Photon Fusion 2, or fully standalone with no networking in the scene. The core assembly builds against an empty reference list, so the same behaviour serves both without a port. Released with public documentation.
- **Built and published *claude-code-unity-kit*, a Claude Code plugin for Unity 6** (MIT, v0.2.0) — deterministic guardrails around agent work — the turn cannot close while the project has compile errors, an orphaned *.meta*, or a reference broken since the last commit, and "not checked" stays a distinct result from "passed" so the gate never launders uncertainty into a green tick. Reference checking rewritten 18s → 4.6s on a 3,000-asset project. · github.com/akaashg/claude-code-unity-kit
- **Designed *xrapp.dev*, a multi-tenant VR training platform where the organisation is the tenant** — entitlement, assignment and attempt modelled separately so training history survives roster and catalogue changes. Nine-table PostgreSQL model behind a Next.js admin with Drizzle and Supabase Auth, Stripe billing scoped to the team rather than the user, and a separate containerised service bridging progress back to the customer's LMS.
- **Built *Ruins of the Forgotten*, an open-world survival game with a public Steam demo** (Early Access planned Q3 2026) — procedurally generated jungle islands, day/night simulation that turns wildlife hostile after dark, animation state machines, custom HLSL shaders, and the full Steamworks pipeline through build, depot, store page and age-rating submission. · store.steampowered.com/app/2726410
- Delivered **15+ XR applications** across Meta Quest, HoloLens, HTC Vive and Varjo — training simulators, multiplayer experiences and AR visualization using passthrough MR, hand tracking and room-scale interaction — plus a gamified VR compliance product that turned regulatory curricula into branching, scored scenarios. Each held to budget through draw-call batching, LOD, texture atlasing, GC-aware pooling and custom URP/HDRP shaders.

Lead VR Developer · Witzeal Technologies

Mar 2023 – Jul 2023 · Contract · Remote

Multiplayer Meta Quest extension of Big Cash Poker — a 50M+ user money-handling platform.

- Designed the integration between a Unity VR client and the client's **existing proprietary Java backend, which could not be changed** — so the headset became a view and never an authority. Held **sub-100ms** across the service boundary through optimized state synchronization and authoritative server reconciliation.
- Led end-to-end Unity development of the multiplayer experience — real-time multi-user game logic, networked interaction and 3D environment rendering.
- Established sprint cadence and code review — shipped a regulated, money-handling multiplayer product with **zero critical post-launch defects**, 25% inside a fixed three-month window.

Lead Unity Developer · CodeXAR

Jan 2022 – Dec 2022 · Remote

CollaboratAR — enterprise multi-user AR design review across geographies.

- Led a **5-person team** building CollaboratAR in Unity with AR Foundation, Mirror Networking, Master Server Toolkit and Draco mesh compression — letting distributed stakeholders review detailed 3D clones of physical products together, on web and in AR.
- Built the **photogrammetry capture-to-runtime pipeline**, holding **8K source texture detail** on mobile hardware in real time through shader optimization, texture atlasing, Draco compression and LOD — a virtual sample nobody trusts replaces nothing.
- Cut clients' physical sampling and travel costs **40%**, validated by their procurement metrics. Shipped to iOS and Android: build pipelines, store certification and post-launch QA.

Founder & Lead Developer · Cyber Cerebs

2009 – 2021 · Delhi, India

Digital marketing agency and software development company.

- Grew a solo practice into an **8-person team** delivering **50+ products** for clients across hospitality, sports, travel and logistics in India and overseas — websites and web applications, mobile apps, marketing automation, and the REST APIs, relational schemas and third-party integrations behind them.
- Owned architecture, technical hiring, client engagement, concurrent delivery timelines and P&L across a multi-vertical portfolio for twelve years.

SELECTED WORK

Shipped · *Photon Enemy AI* v1.0.0 (Unity package, public docs) · *claude-code-unity-kit* MIT v0.2.0 · CollaboratAR (enterprise AR, 40% client cost reduction) · Big Cash Poker VR (50M+ user IP)

In development · LBE VR Arena — pre-deployment, 8-player co-located · *Ruins of the Forgotten* — public Steam demo live, Early Access Q3 2026 · *xrapp.dev* — multi-tenant training platform

Independent mobile titles · *Rush Hour* (1K+ downloads), *Hide and Seek* (500+), *ABS* — live on Google Play, maintained through IL2CPP/ARM64 and 16KB page-size requirements

Links · store.steampowered.com/app/2726410 · github.com/akaashg/claude-code-unity-kit · 2-dude-studios.github.io/Enemy-AI-Photon-Fusion-Docs · case studies at · akaashgarg.com/projects

TECHNICAL SKILLS

Architecture: Distributed state and authority models, service boundaries, coordinate and reference frames, relational schema design, offline-first sync with retry and reconciliation, failure and recovery design, multi-tenant models, architecture decision records

AI-Native & Agentic Engineering: Agent guardrails and verification gates, context engineering (CLAUDE.md and project rule files), Claude Code plugins, skills, hooks and review subagents, A/B evals for agent behaviour, deterministic checks around non-deterministic work

Real-Time & Multiplayer: Photon Fusion 2, Photon PUN, Photon Voice, Mirror Networking, server-authoritative state, client-side prediction & interpolation, reconciliation, latency compensation, dedicated server hosting

Backend & Data: PostgreSQL, Node.js, Express, Next.js, Drizzle, Supabase, React, REST API design, authentication & session management, Stripe, Python/Django, GitHub Actions, Cloudflare Workers

Engine, Graphics & XR: Unity 6, C#, OpenXR, AR Foundation, XR Interaction Toolkit, Meta XR SDK, MRTK, ARKit/ARCore, Three.js/WebXR, URP/HDRP, HLSL, ShaderGraph, GPU profiling, draw-call batching, LOD, GC-aware pooling, 72fps/13.9ms budgets

Platforms & Tools: Meta Quest 2/3/Pro/3S, HoloLens, HTC Vive, Varjo, Steam/Steamworks, Google Play Console, iOS, Android · C#, TypeScript, Python, SQL, HLSL · Git, Unity Profiler, RenderDoc, Blender, Agile/Scrum